Faithful or Plausible? A Ground-Truth Evaluation of Explainable AI for 3D MRI-Based Alzheimer's Disease Classification

Uday Bhardwaj, Aadarsh Ahlawat, Karamjodh Singh, Arun Kumar Bhagat, Richa Tiwari

Department of Computer Science & Engineering (AI & ML), Chandigarh University, Mohali, Punjab, India

Status of this draft (31 October 2026). Section 5.2 reports the benchmark on the full IXI cohort (581 subjects) with 5-fold cross-validation, every subject explained by all six methods, including the shortcut sweep. The randomisation tests (E6) and the Occlusion window ablation (E7) use 100 subjects (20 per fold). The remaining ablations of E7 are not run. Every benchmark number is generated from the analysis output; none is projected or invented.

Abstract

Explanation heatmaps for deep-learning models of Alzheimer's disease (AD) are commonly validated by showing that they highlight the hippocampus and medial temporal lobe. Such anatomical plausibility does not establish faithfulness, i.e. that the highlighted voxels drive the model's decision, and on real cohorts neither property can be checked against ground truth. We present a benchmark in which the disease signal is known voxel by voxel. Alzheimer-like hippocampal atrophy is simulated on healthy T1-weighted MRI from the open IXI dataset by a smooth, fold-checked displacement field whose Jacobian provides an exact ground-truth map; a faint intensity shortcut with controlled label correlation is optionally planted far from the hippocampus. Every scan, including controls, is resampled through the same warp so that interpolation cannot act as a label cue. We evaluate six attribution methods (LRP with BatchNorm folding, Grad-CAM, Grad-CAM++, Integrated Gradients, Occlusion and Gradient × Input) for a 3D ResNet on three axes: plausibility (atlas-region relevance), faithfulness (insertion/deletion AUC) and correctness (agreement with ground truth). During implementation, a relevance-conservation test exposed a sign error in the LRP input-layer rule that produced plausible-looking but incorrect maps; correcting it moved LRP from below to above a random-map faithfulness baseline on a synthetic model. In five-fold cross-validation on 581 IXI subjects, the classifier detected simulated atrophy (test AUC 0.992 ± 0.006) and the planted shortcut (0.826 ± 0.070), and stayed at chance on the resampling control (0.515 ± 0.036). On atrophy, the six methods ranked identically on faithfulness and on correctness, with Occlusion first and LRP the best gradient-based method, whereas plausibility exaggerated their differences. Faithfulness scores on the control model, which had learned nothing, reached about three-quarters of those on the atrophy model, so faithfulness alone does not show that a map explains a meaningful decision. A location control showed that Grad-CAM and Grad-CAM++ assign 2.7–3.0 times chance relevance to the shortcut's position in a model that never saw a shortcut; without it they would be credited with detecting a shortcut that the model, by an explanation-independent flip test, barely uses. LRP, Integrated Gradients and Gradient × Input attributed relevance to the shortcut in proportion to the model's actual dependence on it, and when a stronger shortcut made the model abandon atrophy every method followed it, while hippocampal plausibility fell below chance. In randomisation tests, re-initialising the output layer left Grad-CAM++ maps almost unchanged (Spearman $\rho = 0.95$) and Gradient × Input decorrelated only partly, so these two maps reflect the input more than the model. Per-volume correlations between plausibility and faithfulness were at most moderate ($|\rho| \leq 0.54$) and changed sign between conditions. Evaluating explanations needs ground truth, a reliance test and a no-signal control; plausibility is no substitute.

Keywords: explainable AI, layer-wise relevance propagation, faithfulness, Alzheimer's disease, structural MRI, ground truth, shortcut learning

1. Introduction

Convolutional networks classify AD from structural MRI with high reported accuracy, and attribution maps are increasingly presented as evidence that these models attend to disease-relevant anatomy [6, 8]. The usual argument is visual: relevance concentrates on the hippocampus, which is what a clinician expects. Jacovi and Goldberg [12] distinguish this plausibility from faithfulness, the degree to which an explanation reflects the model's actual reasoning, and argue that the two must be evaluated separately.

For AD the separation matters for three reasons. Models exploit shortcuts such as age, scanner and acquisition artefacts [9]; a faithful explanation of such a model should reveal the shortcut, not the hippocampus. Data leakage between visits of the same subject inflates accuracy [20], so explanations of leaky models are uninterpretable. And attribution implementations can be wrong while still producing convincing maps, as we show for LRP in Section 5.

The obstacle is ground truth. On a real cohort nobody knows which voxels a model should use, so a plausible map cannot be shown to be unfaithful. We remove this obstacle by constructing the disease signal ourselves on healthy brains. Our contributions are:

- **A ground-truth benchmark for 3D MRI explanations**, built on open IXI data, with simulated hippocampal atrophy, an exact voxel-wise truth map, a planted shortcut with controlled label correlation, and a resampling control.
- **A three-axis evaluation** of six attribution methods: plausibility, faithfulness and correctness, measured on the same maps.
- **Controls that change the conclusions:** a reliance test that needs no explanation, a sweep over shortcut strength, a no-signal location control and model- and data-randomisation tests.
- **Implementation validity checks** for LRP in residual 3D networks: relevance conservation, BatchNorm folding equivalence and agreement with a reference implementation, including a documented sign error that conservation testing caught.

2. Related Work

Attribution methods. LRP [4, 14] redistributes the output score backwards under a conservation constraint, with composite rules recommended for deep networks. Grad-CAM [17] weights the final convolutional feature maps by pooled gradients; in 3D it is upsampled from a coarse grid. Integrated Gradients [18] integrates gradients along a path from a baseline. Occlusion and Gradient × Input serve as model-agnostic and first-order baselines.

Explanations for AD. Böhle et al. [6] applied LRP to a 3D CNN on ADNI and found relevance in the hippocampus and temporal lobe. Dyrba et al. [8] evaluated relevance maps for AD, including a simulated-atrophy check in which intensity was reduced in a cube. Studies of this kind establish plausibility; faithfulness is rarely measured on the same maps.

Evaluating explanations. Insertion/deletion curves [15] measure how the score changes as features are added or removed in order of attribution. ROAR retrains after removal, which is costly in 3D, and ROAD

[16] reduces leakage from the removal pattern. Sanity checks [1] test whether maps depend on the model at all. In medical imaging, Arun et al. [3] found that many saliency methods fail localisation and randomisation tests on chest radiographs.

Simulated atrophy. Biophysical simulators generate longitudinal MRI with known volume change [13]. Generative models produce realistic synthetic brains but without voxel-level ground truth for a planted effect. We use a lightweight displacement-field model chosen for an exact, verifiable truth map.

3. Methods

3.1 Data and Preprocessing

We use the IXI dataset [11]: approximately 580 T1-weighted scans of healthy adults aged 20–86 from three London sites, released under CC BY-SA 3.0. Each scan is bias-corrected (N4), affinely registered head-to-head to the ICBM152 2009 template on the 2 mm MNI grid, brain-extracted with the template brain mask (dilated by two voxels), resampled to 96^3 and z-scored within the brain mask. A bilateral hippocampus mask is taken from the Harvard-Oxford subcortical atlas and resampled identically; its placement was checked visually on scans from two sites. All 581 scans were processed without failure. The cohort is 48.7 ± 16.5 years old (range 20–86; 18 ages missing), 313 female and 253 male (15 missing), from Guys (322), Hammersmith (185) and the Institute of Psychiatry (74).

3.2 Simulated Atrophy with Exact Ground Truth

Let c_k be the centroid of hippocampal component k (left, right), $q_k(x)$ the nearest point of component k to voxel x, and $d_k(x)$ the distance to it. We define the backward displacement $u(x) = s \Sigma_k w_k(x)(q_k(x) - c_k)$, with $w_k = 1$ inside the component and $w_k = \exp(-d_k^2/2\sigma^2)$ outside, and resample $\text{out}(x) = \text{in}(x + u(x))$. Inside each component the field is a uniform shrink by $1/(1+s)$, so a star-shaped structure loses a fraction $1 - (1+s)^{-3}$ of its volume; outside, the boundary displacement fades with distance. The scale s is set from a target volume loss drawn uniformly from 15–30%. The ground-truth map is $|\det(I + \nabla u) - 1|$, thresholded at 0.02. A deformation with non-positive Jacobian determinant anywhere is rejected. The achieved volume loss is measured per subject by partial-volume integration of the warped mask. Across the 291 positives of the cohort, the achieved loss was 21.2% on average (range 13.8–28.7%) for a mean target of 22.6%, and the smallest Jacobian determinant was 0.605, so no deformation folded.

3.3 Resampling Control

Warping interpolates and smooths. To prevent a classifier from detecting interpolation rather than atrophy, every scan, positive or control, is resampled once through a random smooth sub-voxel jitter field (maximum 0.5 voxel), to which the atrophy field is added for positives. The jitter is excluded from the ground truth.

3.4 Planted Shortcut and Benchmark Conditions

The shortcut is a Gaussian intensity bump (amplitude 0.3 in z-score units, $\sigma = 4$ voxels), truncated at 5% of its peak so that no voxel outside its ground-truth support changes, and placed at the brain voxel farthest from the hippocampus. Labels are balanced within acquisition site and across age. Three conditions are built: A (atrophy only), B (atrophy, with the shortcut present in exactly a fraction p of

positives and 1 − p of negatives, p ∈ {0.7, 0.9, 1.0}, and at p = 0.9 also with the bump's amplitude raised to 1.0) and C (shortcut only). For each shortcut condition a flipped test set moves the shortcut to the opposite class, giving a direct, explanation-independent measure of reliance.

3.5 Classifier

A 3D ResNet-10 at half width is trained with subject-level 5-fold cross-validation, class-weighted loss, mixed precision and early stopping on validation AUC. A split-integrity check raises an error if any subject appears in more than one partition.

3.6 Attribution Methods

All methods explain the model's predicted class. LRP uses a composite of z^B (input layer), α1β0 (lower layers) and ε (upper layers); residual blocks are propagated as single units so that relevance is split between the main and skip branches, and BatchNorm is folded into the preceding convolutions before propagation. Occlusion replaces a 16^3 window, slid at stride 8, by zero (the brain's mean intensity after z-scoring) and records the drop in the class probability; Integrated Gradients uses 32 steps from a Gaussian-blurred copy of the scan (σ = 5 voxels).

3.7 Evaluation Axes

- **Plausibility:** share of positive relevance inside the bilateral hippocampus atlas mask, reported as enrichment, i.e. divided by the mask's share of brain volume (1.0 = no concentration).
- **Faithfulness:** insertion AUC minus deletion AUC over 8^3 patches with a shared blurred baseline, compared against a random map.
- **Correctness:** AUPRC of the map against the atrophy and shortcut ground-truth supports, reported as lift, i.e. AUPRC divided by its chance level (the support's share of the brain), so that 1.0 means no better than a random map whatever the size of the target.

Statistics are computed over subjects, which contribute one scan each; 95% confidence intervals are percentile bootstraps over subjects (2,000 resamples).

4. Experimental Design

#	Experiment	Tests	Status
E1	Classification on conditions A, B, C (subject-level 5-fold CV)	Signal learned	Done (5 folds)
E2	Controls vs controls through the shared warp	Resampling cue	Done: at chance
E3	Six methods × three axes on condition A	H3, H5	Done (581 subjects)
E4	Shortcut sweep over p and amplitude, with flipped test sets	H2	Done (581 subjects per set)
E5	Plausibility–faithfulness–correctness divergence	H1	Done (581 subjects per set)
E6	Cascading and data randomisation	H4	Done (100 subjects)
E7	Ablations: Occlusion window; baseline, LRP split, severity	Robustness	Window done (100 subjects); others pending

#	Experiment	Tests	Status
E8	Conservation, BatchNorm folding, Zennit agreement	Validity	Partly done (Section 5)
E9	Shortcut-location control on a model without shortcut (added after pilot)	Location bias	Done (exploratory)

Table 1. Experiments

Hypotheses, fixed before results: **H1** plausibility and faithfulness are weakly correlated per volume ($|\rho| < 0.3$); **H2** on shortcut-reliant models at least one method retains high hippocampal plausibility with low correctness against the shortcut; **H3** Grad-CAM scores relatively higher on plausibility than correctness, LRP the reverse; **H4** methods differ in decorrelation under weight randomisation; **H5** in condition A, faithfulness rankings agree with correctness rankings. Statistics are computed over subjects with bootstrap confidence intervals and Holm correction.

5. Results

5.1 Verified Implementation Results

Check	Before fix	After fix
LRP conservation, relative error (bias-free model)	1.97	1.3e-5
LRP faithfulness score, synthetic model (random: −0.007)	−0.014	+0.035
BatchNorm fold, max |output difference|	–	9.8e-7
Atrophy: achieved vs 25% target volume loss (phantom)	13.9% (blurred weight)	22.9%
Ground-truth mass within 2σ of structure (phantom)	61% (centroid field)	92.8%
Shortcut: voxels changed outside its ground truth	Non-zero (untruncated tail)	0

Table 2. Implementation checks on synthetic data (last run 22 October 2026)

The LRP sign error is instructive. The z^B input rule computes $z = \text{conv}(a, W) - \text{conv}(l, W^{+}) - \text{conv}(h, W^{-})$; the gradients with respect to the bounds l and h already carry the minus sign, so the relevance must be $a \cdot \nabla a + l \cdot \nabla l + h \cdot \nabla h$. The implementation subtracted the bound terms, so input relevance summed to roughly minus the logit. The heatmaps remained visually plausible. Only the conservation test revealed the error, and correcting it moved LRP from below to above the random-map faithfulness baseline. Two simulator variants were likewise rejected by tests, as shown in the table.

5.2 Benchmark Results

All results use subject-level 5-fold cross-validation: in each fold about 371 subjects train, 93 validate and 116 test, and every subject is tested exactly once, so classification and attribution metrics cover all 581 subjects. The flipped test sets contain the same subjects with the shortcut moved to the opposite class. Confidence intervals are bootstraps over subjects pooled across folds.

5.2.1 Learned Signal and Resampling Control (E1, E2)

Condition	Test AUC, mean ± sd over 5 folds	Test accuracy	Best epoch	Accuracy, shortcut flipped	Logit shift towards the shortcut's class
Resampling control	0.515 ± 0.036	0.484	0–26	–	–
A: atrophy only	0.992 ± 0.006	0.943	10–49	–	–
C: shortcut only	0.826 ± 0.070	0.747	16–41	0.253	6.46 [6.19, 6.74]
B: atrophy + shortcut, p = 0.9	0.980 ± 0.025	0.924	12–24	0.921	0.10 [0.09, 0.11]

Table 3. Classification (581 subjects, 5 folds) and shortcut reliance

The classifier detected simulated atrophy almost perfectly and the planted shortcut less reliably, consistent with the bump's low amplitude (0.3 standard deviations of brain intensity). On the resampling control, where labels were assigned but no signal was planted, the test AUC was 0.515 ± 0.036, with every fold between 0.47 and 0.56, and test accuracy 0.484. The shared warp therefore does not leak the label.

5.2.2 Shortcut Reliance (E4)

Moving the shortcut to the opposite class reduced the shortcut-only model's accuracy from 0.747 to 0.253, and shifted its logit by 6.46 towards the class the bump indicates: the model relies on the bump. When the same bump accompanied atrophy with p = 0.9, accuracy barely changed (0.924 with the shortcut in place, 0.921 flipped), 0.4% of decisions changed and the logit moved by 0.10, 1.5% of the shortcut-only model's shift. That model uses atrophy and depends on the shortcut only marginally, although it saw the shortcut in 90% of matching training scans. Because reliance is measured without any explanation, the two models are reference cases against which the maps can be judged.

5.2.3 Three Axes on Atrophy (E3)

Method	Faithfulness (ins − del)	Plausibility (enrichment)	Correctness: atrophy (lift)
LRP	0.617 [0.595, 0.638]	2.14 [2.06, 2.23]	1.82 [1.77, 1.86]
Grad-CAM	0.301 [0.276, 0.327]	1.14 [1.12, 1.16]	1.32 [1.26, 1.39]
Grad-CAM++	0.292 [0.265, 0.318]	1.14 [1.13, 1.15]	1.25 [1.20, 1.30]
Integrated Gradients	0.558 [0.533, 0.583]	1.40 [1.36, 1.43]	1.58 [1.55, 1.60]
Gradient × Input	0.546 [0.521, 0.571]	1.27 [1.23, 1.30]	1.57 [1.55, 1.60]
Occlusion	0.859 [0.848, 0.871]	6.09 [5.80, 6.38]	3.17 [3.03, 3.31]
Random map	0.013 [−0.013, 0.039]	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]

Table 4. Condition A, 581 subjects: mean [95% bootstrap CI over subjects]

All six methods exceeded the random map on faithfulness. Ranked by faithfulness and by correctness against the atrophy ground truth, the methods fall in the same order: Occlusion, LRP, Integrated Gradients, Gradient × Input, Grad-CAM, Grad-CAM++ (Spearman ρ = 1.0, exact permutation p = 0.003), although Integrated Gradients and Gradient × Input are tied within their intervals on both axes. H5 is supported. Plausibility, in contrast, exaggerated the differences: Occlusion concentrated 6.1 times chance relevance on the hippocampus and LRP 2.1 times, against correctness of 3.2 and 1.8. H3 is not supported:

Grad-CAM scored lower on plausibility than on correctness (1.14 against 1.32), and LRP the reverse (2.14 against 1.82). Correctness was modest for the gradient-based methods (1.25–1.82 times chance): the atrophy ground truth includes the displaced tissue around the hippocampus, and every map also places relevance elsewhere.

5.2.4 A Model That Uses the Shortcut (Condition C)

Method	Faithfulness (ins – del)	Plausibility (enrichment)	Correctness: shortcut (lift)
LRP	0.636 [0.616, 0.656]	0.22 [0.22, 0.23]	17.62 [16.35, 18.99]
Grad-CAM	0.346 [0.321, 0.371]	0.79 [0.78, 0.81]	7.36 [6.58, 8.23]
Grad-CAM++	0.322 [0.297, 0.345]	0.81 [0.80, 0.82]	8.19 [7.36, 9.04]
Integrated Gradients	0.584 [0.561, 0.607]	0.19 [0.18, 0.20]	8.17 [7.53, 8.85]
Gradient × Input	0.572 [0.548, 0.595]	0.16 [0.15, 0.17]	8.27 [7.63, 8.97]
Occlusion	0.883 [0.874, 0.891]	0.18 [0.16, 0.20]	29.62 [27.62, 31.75]
Random map	−0.030 [−0.061, 0.001]	1.00 [1.00, 1.00]	1.01 [1.00, 1.01]

Table 5. Condition C, 581 subjects: mean [95% bootstrap CI over subjects]

All methods located the shortcut well above chance, from 7.4 times chance for Grad-CAM to 29.6 for Occlusion, and all placed less than chance relevance on the hippocampus (enrichment 0.16–0.81). No method produced a plausible hippocampal map for this model, so H2 is not supported: on a model that relied on the shortcut, low plausibility correctly signalled the problem.

5.2.5 A Model That Barely Uses the Shortcut, and a Location Control (Condition B, p = 0.9)

Method	Faithfulness (ins – del)	Plausibility (enrichment)	Correctness: atrophy (lift)	Correctness: shortcut (lift)
LRP	0.558 [0.529, 0.589]	2.21 [2.11, 2.31]	1.67 [1.63, 1.73]	1.33 [1.27, 1.39]
Grad-CAM	0.300 [0.271, 0.332]	1.15 [1.14, 1.17]	1.34 [1.29, 1.40]	2.53 [2.25, 2.84]
Grad-CAM++	0.301 [0.271, 0.332]	1.17 [1.16, 1.18]	1.25 [1.21, 1.30]	2.67 [2.45, 2.90]
Integrated Gradients	0.505 [0.473, 0.538]	1.40 [1.36, 1.44]	1.45 [1.43, 1.48]	1.07 [1.05, 1.08]
Gradient × Input	0.498 [0.466, 0.532]	1.28 [1.25, 1.31]	1.47 [1.44, 1.50]	1.01 [1.00, 1.03]
Occlusion	0.802 [0.786, 0.820]	4.90 [4.62, 5.18]	2.57 [2.45, 2.69]	3.41 [3.03, 3.81]
Random map	0.007 [−0.016, 0.030]	1.00 [1.00, 1.00]	1.00 [1.00, 1.00]	1.01 [1.00, 1.01]

Table 6. Condition B with p = 0.9, 581 subjects: mean [95% bootstrap CI over subjects]

On this model Grad-CAM, Grad-CAM++ and Occlusion assigned the shortcut 2.5–3.4 times chance relevance. The shortcut sits at the same position in every scan, so a high score against its ground truth can come from the bump or from a preference for that position. To separate the two we explained the atrophy-only model, which never saw a bump, against the same location mask (E9; this control was added after the pilot and was not among the pre-specified experiments). The excess of a method's lift in B over

this baseline, as a share of its excess in C, estimates how much of its shortcut relevance in C it retains on a model that barely uses the shortcut; the model itself retains 1.5% of the logit dependence.

Method	A: no bump (location baseline)	B: bump barely used	C: bump used	B excess over baseline, % of C excess
LRP	0.87 [0.87, 0.88]	1.33 [1.27, 1.39]	17.62 [16.35, 18.99]	+2.7%
Grad-CAM	2.68 [2.48, 2.87]	2.53 [2.25, 2.84]	7.36 [6.58, 8.23]	−3.2%
Grad-CAM++	3.04 [2.83, 3.26]	2.67 [2.45, 2.90]	8.19 [7.36, 9.04]	−7.3%
Integrated Gradients	0.90 [0.90, 0.91]	1.07 [1.05, 1.08]	8.17 [7.53, 8.85]	+2.3%
Gradient × Input	0.88 [0.87, 0.88]	1.01 [1.00, 1.03]	8.27 [7.63, 8.97]	+1.8%
Occlusion	0.93 [0.91, 0.94]	3.41 [3.03, 3.81]	29.62 [27.62, 31.75]	+8.7%
Random map	1.01 [1.00, 1.01]	1.01 [1.00, 1.01]	1.01 [1.00, 1.01]	–

Table 7. Lift against the shortcut location, 581 subjects per condition

Grad-CAM and Grad-CAM++ assigned 2.7 and 3.0 times chance relevance to the shortcut's position in a model that had never seen a shortcut, as much as in condition B. Their apparent detection of the shortcut on the B model is therefore a property of the position, not of the bump, and even on the shortcut-reliant model their lift exceeds this baseline by less than a factor of three, against a factor of about twenty for LRP. Without the location control, a ground-truth evaluation would credit them with finding a shortcut that the model barely uses. LRP, Integrated Gradients and Gradient × Input placed slightly less than chance relevance at that position without a bump and retained 1.8–2.7% of their condition-C excess in B, in proportion to the model's residual dependence; Occlusion retained 8.7%, overstating that dependence about sixfold.

5.2.6 Plausibility Versus Faithfulness (E5)

Method	Resampling control	A: atrophy only	C: shortcut only	B: atrophy + shortcut, p = 0.9
LRP	−0.36	+0.28	+0.25	+0.44
Grad-CAM	−0.05	+0.32	−0.41	+0.33
Grad-CAM++	−0.39	+0.34	−0.50	+0.39
Integrated Gradients	−0.25	+0.16	−0.24	+0.15
Gradient × Input	−0.25	−0.02	+0.19	−0.12
Occlusion	−0.49	+0.37	+0.18	+0.54
Random map	+0.01	+0.04	−0.04	+0.02

Table 8. Spearman ρ between per-volume plausibility (hippocampus mass) and faithfulness, 581 subjects per condition

H1 is partly supported. Plausibility and faithfulness were never strongly correlated ($|\rho| \leq 0.54$), but $|\rho| < 0.3$ held in only 12 of 24 method–condition pairs, and the sign changed with what the model used. On the control model all methods showed negative correlations: the more a map concentrated on the hippocampus, the less faithful it was. On atrophy models most correlations were positive, and on the

shortcut model Grad-CAM and Grad-CAM++ were again negative (ρ = −0.41 and −0.50). Plausibility tracks faithfulness weakly, and only when the model uses the anatomically expected region.

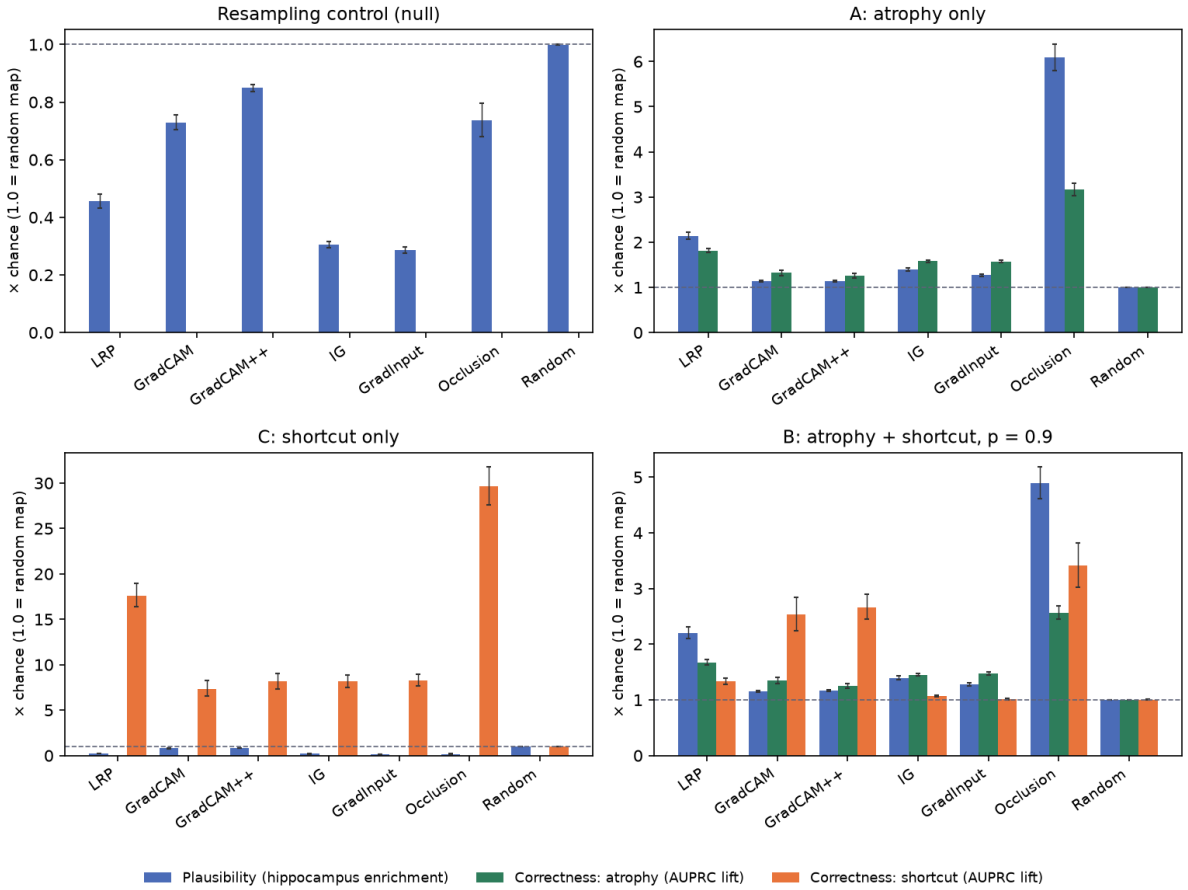


Figure 1. Plausibility (hippocampus enrichment) and correctness (AUPRC lift against the atrophy and shortcut ground truth) per method and condition, each relative to a random map (dashed line at 1.0). Error bars: 95% bootstrap CI over 581 subjects. Shortcut lift in condition B should be read against the location baseline in Table 7.

5.2.7 Faithfulness on a Model That Learned Nothing

On the resampling-control model, which learned nothing about the label, Occlusion, LRP, Integrated Gradients and Gradient × Input reached faithfulness scores of 0.70, 0.47, 0.41 and 0.40, against −0.01 for a random map about three-quarters of their scores on the atrophy model. The insertion–deletion score measures how the logit responds to the highlighted voxels, not whether those voxels carry a label-related signal, so a high faithfulness score alone does not show that a map explains a meaningful decision.

5.2.8 Shortcut Strength (E4)

Condition B	Test AUC, mean ± sd	Accuracy: shortcut in place → flipped	Logit shift towards the shortcut's class	Decisions changed
p = 0.7	0.991 ± 0.006	0.916 → 0.916	0.03 [0.03, 0.03]	0.0%
p = 0.9	0.980 ± 0.025	0.924 → 0.921	0.10 [0.09, 0.11]	0.4%
p = 1.0	0.976 ± 0.026	0.900 → 0.898	0.25 [0.23, 0.26]	0.2%
p = 0.9, amplitude 1.0	0.904 ± 0.025	0.781 → 0.224	8.23 [7.87, 8.61]	69.9%
C: shortcut only (reference)	0.826 ± 0.070	0.747 → 0.253	6.46 [6.19, 6.74]	49.4%

Table 9. Shortcut sweep: classification and reliance, 581 subjects per set (bump amplitude 0.3 unless stated)

Method	Location baseline (A)	p = 0.7	p = 0.9	p = 1.0	p = 0.9, amplitude 1.0
LRP	0.87	0.99	1.33	2.46	26.81
Grad-CAM	2.68	2.69	2.53	3.90	6.07
Grad-CAM++	3.04	3.06	2.67	5.26	7.12
Integrated Gradients	0.90	0.96	1.07	1.34	13.28
Gradient × Input	0.88	0.91	1.01	1.26	13.25
Occlusion	0.93	1.21	3.41	11.82	40.22

Table 10. Shortcut sweep: correctness against the shortcut (AUPRC lift, mean over 581 subjects)

At the original amplitude the model preferred atrophy at every correlation: flipping the shortcut changed at most 0.4% of decisions, even when the bump accompanied every positive and no negative (p = 1.0). Its dependence on the bump nevertheless grew with p, the logit shift rising from 0.03 to 0.10 to 0.25. LRP, Integrated Gradients, Gradient × Input and Occlusion followed this small rise monotonically (shortcut lift 0.99 → 1.33 → 2.46, 0.96 → 1.07 → 1.34, 0.91 → 1.01 → 1.26 and 1.21 → 3.41 → 11.82 as p went from 0.7 to 1.0). Grad-CAM and Grad-CAM++ stayed at their location baseline (2.68 and 3.04) until p = 1.0, so on these models their shortcut scores carried no information about reliance.

Raising the bump's amplitude to 1.0 produced a model that switched to the shortcut: flipping it reduced accuracy from 0.781 to 0.224 and changed 70% of decisions, and the logit shift (8.23) exceeded that of the shortcut-only model. Every method followed the switch. Shortcut lift rose to 40.2 for Occlusion and 26.8 for LRP, atrophy lift fell to 0.87–1.00 and hippocampal enrichment to 0.31–0.91, below chance for every method. H2 is again not supported. These maps look wrong by the plausibility criterion, yet they correctly describe a model that has stopped using the hippocampus; only the reliance test and the ground truth show that the maps are right about the model.

5.2.9 Randomisation Tests (E6)

Method	fc	layer4	layer3	layer2	layer1	stem	Data: ρ	Data: SSIM
LRP	−0.25	+0.15	+0.10	+0.11	+0.10	+0.10	0.14	0.71
Grad-CAM	+0.21	+0.37	+0.00	−0.01	−0.05	−0.04	0.10	0.05
Grad-CAM++	+0.95	+0.67	+0.59	+0.70	+0.71	+0.29	0.57	0.50
Integrated Gradients	−0.11	+0.24	+0.15	+0.16	+0.16	+0.15	0.17	0.89
Gradient × Input	+0.17	+0.47	+0.43	+0.43	+0.42	+0.43	0.44	0.89
Occlusion	−0.27	+0.18	+0.12	+0.14	+0.13	+0.14	0.24	0.10

Table 11. Sanity checks, 100 subjects of the atrophy models (20 per fold). Cascading: Spearman ρ of |map| with the original after re-initialising the network from the output layer (fc) down to and including the named stage. Data: similarity to the map of the same scan under the resampling-control model, whose labels are unrelated to the images. Pass: values near zero.

Re-initialising only the output layer already decorrelated the maps of LRP, Integrated Gradients and Occlusion from the original (ρ = −0.25, −0.11 and −0.27), and after further stages their correlation with

the original stayed at 0.10–0.15, 0.15–0.24 and 0.12–0.18: these maps depend on the learned weights. Grad-CAM decorrelated from layer3 on, but 22–49% of its maps were constant once layer4 was randomised and are excluded from ρ, as are 15–40% of Occlusion's from layer3 on: a random network often leaves these methods nothing to show. Grad-CAM++ failed the test. Its map was almost unchanged by a random output layer (ρ = 0.95) and kept 0.59–0.71 through layer1, falling to 0.29 only when the stem was randomised. Gradient × Input decorrelated only partly, to a plateau of 0.42–0.47, because multiplying by the input carries image structure into the map whatever the weights.

Data randomisation agreed. Maps of the same scans under the control model correlated with the atrophy model's maps at ρ = 0.57 for Grad-CAM++ and 0.44 for Gradient × Input, against 0.10–0.24 for the other methods. SSIM was high for Integrated Gradients, Gradient × Input (0.89) and LRP (0.71) despite low rank correlation, because all three maps follow the brain outline; as in the original test [1], rank correlation is the primary measure. H4 is supported: the methods differ in how much their maps depend on the model, and Grad-CAM++ maps largely reflect the input and architecture rather than what the model learned.

5.2.10 Occlusion Window (E7)

Window (stride)	Plausibility (enrichment)	Faithfulness (ins − del)	Correctness: atrophy (lift)	Lift at shortcut location, model A	Correctness: shortcut, model B (lift)
8^3 (4)	4.15 [3.75, 4.57]	0.86 [0.83, 0.89]	3.09 [2.88, 3.30]	0.90 [0.88, 0.93]	1.92 [1.62, 2.23]
16^3 (8) default	6.43 [5.77, 7.08]	0.87 [0.84, 0.89]	2.98 [2.63, 3.33]	0.91 [0.89, 0.94]	3.21 [2.49, 4.00]
32^3 (16)	3.55 [3.29, 3.80]	0.69 [0.64, 0.75]	3.50 [3.08, 3.91]	0.99 [0.90, 1.10]	1.90 [1.41, 2.51]

Table 12. Occlusion at three window sizes, 100 subjects (20 per fold): mean [95% bootstrap CI]

Occlusion's correctness against atrophy hardly depended on the window (3.09, 2.98 and 3.50 at 8^3, 16^3 and 32^3), so its lead over the other methods on that axis is not a product of its coarse maps. Plausibility, in contrast, varied from 3.55 to 6.43 and peaked at the default window: the same model and the same scans give hippocampal concentrations that differ by a factor of almost two depending on a setting of the method. Faithfulness was unchanged between 8^3 and 16^3 and lower at 32^3 (0.69). At no window did Occlusion favour the shortcut's position in the atrophy model (lift 0.90–0.99). On the model that barely uses the shortcut, its shortcut lift was 3.21 at the default window but 1.92 and 1.90 at the others, so much of Occlusion's overstatement of that model's shortcut dependence (Section 5.2.5) comes from the default window.

6. Discussion and Limitations

Findings. The benchmark separates the three axes empirically. Faithfulness and correctness agreed: on atrophy the six methods ranked identically on both. Plausibility did not: it exaggerated differences between methods, was at best moderately correlated with faithfulness, and changed sign between conditions. Faithfulness was in turn necessary but not sufficient, since it was almost as high on a model that had learned nothing as on one that had learned the disease signal. Correctness against ground truth was the only axis that distinguished these cases, and it too needed a control: Grad-CAM and Grad-CAM++ favour a particular brain position strongly enough to appear to detect a shortcut placed there. A benchmark that plants a signal at a fixed location should therefore always report a no-signal baseline at

that location. LRP, once its conservation error was corrected, was the best gradient-based method on atrophy correctness, and together with Integrated Gradients and Gradient × Input attributed relevance to the shortcut in proportion to the model's actual dependence on it, across the whole sweep; when a stronger bump made the model switch to the shortcut, every method followed. The randomisation tests separate the methods further: Grad-CAM++ maps barely depend on the model and Gradient × Input maps only partly, so a method can look reasonable on a trained model while explaining little of it. Finally, plausibility is sensitive to settings of the method itself: changing Occlusion's window altered its hippocampal concentration by a factor of almost two while its correctness stayed the same.

Scope. Each fold was trained once with one seed, so confidence intervals reflect variation between subjects, not between training runs. The sweep produced models that barely use the shortcut or switch to it, but none that uses both signals in comparable measure; amplitudes between 0.3 and 1.0 would be needed for that. The randomisation tests and the window ablation were run on 100 subjects (20 per fold) because of their cost, and the ablations of the IG baseline, the LRP residual split and atrophy severity remain to be run. Occlusion's maps are coarse (16^3 windows at stride 8); the window ablation shows that this affects its plausibility and shortcut scores but not its atrophy correctness. The location control was added after the pilot and is reported as exploratory.

Simulation. Simulated atrophy is not Alzheimer's disease. It reproduces the most consistent structural marker, hippocampal volume loss, but not cortical thinning patterns, ventricular dynamics or disease heterogeneity. Our claims therefore concern explanation methods, for which ground truth is the decisive property, not diagnosis. The atrophy model is phenomenological rather than biophysical, and IXI is acquired at three sites, which we control by balancing labels within site. Conclusions about real-cohort behaviour should be confirmed on clinical data where access allows; an openly available AD dataset [5] provides an optional qualitative check.

7. Conclusion

We present a ground-truth benchmark that makes the faithfulness and correctness of 3D MRI explanations measurable rather than assumed. Implementation checks showed that an attribution map can look plausible while being wrong, and the benchmark showed the same at the level of methods. Faithfulness and correctness agreed on how to rank six methods, plausibility did not, and faithfulness was high even for a model that had learned nothing. Grad-CAM and Grad-CAM++ favour a fixed brain position strongly enough to appear to detect a shortcut placed there, which only a no-signal control revealed; LRP, Integrated Gradients and Gradient × Input tracked the model's actual dependence on the shortcut across a sweep of its strength, and every method followed a model that switched to it. Randomisation tests showed that Grad-CAM++ maps barely depend on the model. Plausible-looking maps are not evidence of correct ones; ground truth, a reliance test, a no-signal control and randomisation tests together are.

Acknowledgements

This work uses the IXI dataset (brain-development.org), released under CC BY-SA 3.0.

References

[1] Adebayo, J. et al. (2018). Sanity checks for saliency maps. NeurIPS.

[2] Anders, C. J. et al. (2021). Software for dataset-wide XAI: from local explanations to global insights with Zennit, CoRelAy, and ViRelAy. arXiv:2106.13200.

[3] Arun, N. et al. (2021). Assessing the trustworthiness of saliency maps for localizing abnormalities in medical imaging. Radiology: Artificial Intelligence, 3(6), e200267.

[4] Bach, S. et al. (2015). On pixel-wise explanations for non-linear classifier decisions by layer-wise relevance propagation. PLoS ONE, 10(7), e0130140.

[5] Besga, A. et al. (2020). Alzheimer's disease versus bipolar disorder versus health control MRI data. Zenodo, record 3935636.

[6] Böhle, M. et al. (2019). Layer-wise relevance propagation for explaining deep neural network decisions in MRI-based Alzheimer's disease classification. Frontiers in Aging Neuroscience, 11, 194.

[7] Collins, G. S. et al. (2024). TRIPOD+AI statement. BMJ, 385, e078378.

[8] Dyrba, M. et al. (2021). Improving 3D convolutional neural network comprehensibility via interactive visualization of relevance maps: evaluation in Alzheimer's disease. BMC Medical Imaging, 21, 178.

[9] Geirhos, R. et al. (2020). Shortcut learning in deep neural networks. Nature Machine Intelligence, 2, 665–673.

[10] Hooker, S. et al. (2019). A benchmark for interpretability methods in deep neural networks. NeurIPS.

[11] IXI Dataset. Imperial College London. brain-development.org/ixi-dataset.

[12] Jacovi, A. & Goldberg, Y. (2020). Towards faithfully interpretable NLP systems. ACL.

[13] Khanal, B., Ayache, N. & Pennec, X. (2017). Simulating longitudinal brain MRIs with known volume changes and realistic variations in image intensity. Frontiers in Neuroscience, 11, 132.

[14] Montavon, G. et al. (2019). Layer-wise relevance propagation: An overview. In Explainable AI, LNCS 11700, Springer.

[15] Petsiuk, V., Das, A. & Saenko, K. (2018). RISE: Randomized input sampling for explanation of black-box models. BMVC.

[16] Rong, Y. et al. (2022). A consistent and efficient evaluation strategy for attribution methods. ICML.

[17] Selvaraju, R. R. et al. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. ICCV.

[18] Sundararajan, M., Taly, A. & Yan, Q. (2017). Axiomatic attribution for deep networks. ICML.

[19] Tejani, A. S. et al. (2024). Checklist for Artificial Intelligence in Medical Imaging (CLAIM): 2024 update. Radiology: Artificial Intelligence, 6(4), e240300.

[20] Wen, J. et al. (2020). Convolutional neural networks for classification of Alzheimer's disease: Overview and reproducible evaluation. Medical Image Analysis, 63, 101694.